

z-transform.

The z transform of a sequence is denoted as $z(u_n)$ is defined as

$$z(u_n) = \sum_{n=-\infty}^{\infty} u_n z^{-n} = \bar{u}(z)$$

where $\bar{u}(z)$ is the z transform of u_n and z is a complex number.

- Z-transform exists only when the infinite series in ① is convergent.

The sequence u_n is called the inverse z-transform of $\bar{u}(z)$ and is written as

$$z^{-1}(\bar{u}(z)) = u_n.$$

Example: ① Show that $z(a^n) = \frac{z}{z-a} \quad n \geq 0$

$$z(u_n) = \sum_{n=0}^{\infty} u_n z^{-n}$$

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} \left(\frac{a}{z}\right)^n$$

$$= 1 + \frac{a}{z} + \left(\frac{a}{z}\right)^2 + \dots$$

$$= \left(\frac{1}{1 - \frac{a}{z}} \right)$$

$$\begin{aligned} & 1 + x + x^2 + x^3 + \dots \\ &= \frac{1}{1-x} \end{aligned}$$

$$z(a^n) = \frac{z}{z-a}$$

→ Show that z-transform of $z(n^p) = \dots$

$$z(n^p) = -z \frac{d}{dz} [z(n^{p-1})] \quad n \neq 0$$

where p is an positive integer.

$$i) z[n] = \frac{z}{(z-1)^2} \quad ii) z(n^2) = \frac{z^2+2}{(z-1)^3}$$

$$iii) z(n^3) = \frac{z^3 + 4z + 2}{(z-1)^4}$$

$$\textcircled{1} z[2n] = \frac{2}{(z-1)^2} z[n] = \frac{2}{(z-1)^2} \cdot \frac{z}{(z-1)^2} = \frac{2z}{(z-1)^4}$$

$$\textcircled{2} z[-3n] = -3z[n] = \frac{-3z}{(z-1)^2}$$

$$\textcircled{3} z[2n^2] = 2z[n^2] = \frac{2(z^2+2)}{(z-1)^3}$$

$$\textcircled{4} z(-3n^2) = -3z[n^2] = \frac{-3(z^2+2)}{(z-1)^3} = \frac{-3z^2+2}{(z-1)^3}$$

$$\rightarrow i) z[\cos n\theta] = \frac{z^0 - z \cos \theta}{z^2 - 2z \cos \theta + 1}$$

$$ii) z[\sin n\theta] = \frac{z \sin \theta}{z^2 - 2z \cos \theta + 1}$$

$$\text{Ex. ① } z[\cosh az] = \frac{z^2 - z \cosh a}{z^2 + 0z \cosh a + 1}$$

$$\text{② } z[\sinh az] = \frac{z \sinh a}{z^2 + 0z \cosh a + 1}$$

→ Damping rules.

$$\text{① } i) \quad z[a^n u_n] = [z(u_n)]_z \rightarrow \frac{z}{a}$$

$$\text{ii) } z[\bar{a}^n u_n] = [z(u_n)]_z \rightarrow \frac{z}{\bar{a}}$$

$$\text{② } z[a^n n^2] = \frac{az^2 + a^2 z}{(z-a)^3}$$

$$\text{Ex. - } z[\bar{a}^n n^2] = \frac{\bar{a}z^2 + \bar{a}^2 z}{(\bar{a}-z)^3}$$

$$\text{③ } z[a^n \cos n\theta] = \frac{z^2 - az \cos \theta}{z^2 - 2az \cos \theta + a^2}$$

$$\text{④ } z[a^n \sin n\theta] = \frac{az \sin \theta}{z^2 - 2az \cos \theta + a^2}$$

$$\text{⑤ } \text{Find the } z \text{ transform of } \frac{3n - 4 \sin n\pi/4 + 5a}{z[3n - 4 \sin n\pi/4 + 5a]}$$

By linearity property.

$$z(3n) - 4z\left(\sin n\frac{\pi}{4}\right) + z(5a)$$

$$3z(n) - 4z\left(\sin n\frac{\pi}{4}\right) + 5a(z(1))$$

→ Inverse of z-transform.

1) Find the inverse z transform of

$$U(z) = \frac{2z^2 + 3z}{(z+2)(z-4)}$$

$$U(z) = \frac{2z^2 + 3z}{(z+2)(z-4)}$$

$$= \frac{z(2z+3)}{(z+2)(z-4)}$$

$$\frac{Uz}{z} = \frac{2z+3}{(z+2)(z-4)} \longrightarrow \textcircled{1}$$

By partial fraction method, we can write.

$$\frac{2z+3}{(z+2)(z-4)} = \frac{A}{z+2} + \frac{B}{z-4}$$

$$\frac{2z+3}{(z+2)(z-4)} = \frac{A(z-4) + B(z+2)}{(z+2)(z-4)}$$

$$2z+3 = A(z-4) + B(z+2)$$

$$2z+3 = Az$$

$$\text{Put } z=4$$

$$2(4)+3 = A(4-4) + B(4+2)$$

$$8+3 = A(0) + B(6)$$

$$11 = 0 + 6B$$

$$\boxed{B = \frac{11}{6}}$$

$$\text{Put } z=-2$$

$$2(-2)+3 =$$

$$A(-2-4) + B(-2+2)$$

$$-4+3 = A(-6) + 0$$

$$-1 = -6A$$

$$A = \frac{1}{6}$$

$$\frac{2z+3}{(z+2)(z-4)} = \frac{1/6}{(z+2)} + \frac{11/6}{(z-4)} \rightarrow (2)$$

Put eq (2) in (1).

$$\frac{u(z)}{z} = \frac{1/6}{z+2} + \frac{11/6}{(z-4)}$$

$$u(z) = z \left[\frac{1/6}{z+2} + \frac{11/6}{(z-4)} \right]$$

$$u(z) = \frac{1}{6} \left[\frac{z}{z+2} \right] + \frac{11}{6} \left[\frac{z}{z-4} \right]$$

Take inverse on Both sides.

$$z^{-1}[u(z)] = z^{-1} \left[\frac{1}{6} \left[\frac{z}{z+2} \right] + \frac{11}{6} \left[\frac{z}{z-4} \right] \right]$$

$$u_n = \frac{1}{6} z^{-1} \left[\frac{z}{z+2} \right] + \frac{11}{6} z^{-1} \left[\frac{z}{z-4} \right]$$

we know that
 $z^{-1} \left[\frac{z}{z-a} \right] = a^n$.

$$\therefore u_n = \frac{1}{6} \left[-2(-2)^n + \frac{11}{6} (4)^n \right]$$

2. $U(z) = \frac{5z}{(z-2)(3z-1)}$

$U_2 = \frac{5z}{(z-2)(3z-1)} \longrightarrow \text{⊕}$

$U_2 = \frac{z [5]}{(z-2)(3z-1)}$

$\frac{U_2}{z} = \frac{5}{(z-2)(3z-1)} \longrightarrow \text{ⓐ}$

By partial fraction we know that-

$$\frac{5}{(z-2)(3z-1)} = \frac{A}{(z-2)} + \frac{B}{(3z-1)}$$

$$\frac{5}{(z-2)(3z-1)} = \frac{A(3z-1) + B(z-2)}{(z-2)(3z-1)}$$

$$5 = A(3z-1) + B(z-2)$$

Put $z = 3z-1$
 $3z = 1$

$$\frac{z+1}{1 \times 3} = \frac{5}{3}$$

Put $z = 1/3$

$$\frac{6-1}{3} = \frac{5}{3}$$

$$5 = A(3 \times \frac{1}{3} - 1) + B(2 - 1/3)$$

$$5 = A(0) + B\left(\frac{5}{3}\right)$$

$$z-2$$

$$\underline{\underline{z=2}}$$

$$5 = B\left[\frac{5}{3}\right]$$

$$B = \cancel{5} \times \frac{3}{\cancel{5}} \therefore \boxed{B=3}$$

Put $z = \alpha$.

$$5 = A(3\alpha - 1) + B(\alpha - 2)$$

$$5 = A(3(\alpha) - 1) + B(\alpha - \alpha)$$

$$5 = A(6 - 1) + B(0)$$

$$5 = A(5) + 0$$

$$A(5) = 5$$

$$A = \frac{5}{5} = 1$$

$$\underline{\underline{A = 1}}$$

$$\therefore \frac{5}{(z-2)(3z-1)} = \frac{1}{(z-2)} + \frac{3}{(3z-1)} \rightarrow \textcircled{2}$$

Put eq $\textcircled{2}$ in $\textcircled{1}$.

$$\frac{4z}{z} = \frac{1}{(z-2)} + \frac{3}{(3z-1)}$$

$$\frac{4z}{z} = z \left[\frac{1}{(z-2)} + \frac{3}{(3z-1)} \right]$$

$$\frac{4z}{z} = \frac{1}{(z-2)} + \frac{3z}{z(3z-1)}$$

$$\frac{4z}{z} = \frac{1}{(z-2)} + \frac{1}{(z-1/3)}$$

$$4z = z \left[-\frac{1}{(z-2)} + \frac{1}{(z-1/3)} \right]$$

$$4z = -\frac{z}{(z-2)} + \frac{z}{(z-1/3)}$$

Take inverse on B.S.

$$z^{-1}[U(z)] = z^{-1} \left[\frac{-z}{(z-2)} + \frac{z}{(z-1/3)} \right]$$

$$z^{-1}U(z) = -z^{-1} \left[\frac{z}{(z-2)} \right] + z^{-1} \left[\frac{z}{z-1/3} \right]$$

$$= -2^n + \left(\frac{1}{3}\right)^n \quad \text{B } z^{-1} \left[\frac{z}{z-a} \right] = a^n$$

$$\therefore U_n = -2^n + \left(\frac{1}{3}\right)^n$$

$$\rightarrow U(z) = \frac{182z}{(z-1)(z+1)}$$

$$U(z) = \frac{z(182)}{(z-1)(z+1)}$$

$$\frac{U(z)}{z} = \frac{182}{(z-1)(z+1)} \rightarrow \textcircled{1}$$

By partial fraction

$$\frac{182}{(z-1)(z+1)} = \frac{A}{(z-1)} + \frac{B}{(z+1)}$$

$$\frac{182}{(z-1)(z+1)} = \frac{A(z+1)}{(z-1)(z+1)} + \frac{B(z-1)}{(z-1)(z+1)}$$

$$182 = A(z+1) + B(z-1)$$

$$H^2 + 1$$

$$H^2 = -1$$

$$z = -1/H$$

$$\text{Put } z = -1/H$$

$$\left[\frac{-1}{H} \right] = A \left[\frac{-1}{H} + 1 \right] + B \left[\frac{-1}{H} - 1 \right]$$

$$+ B \left[\frac{-1}{H} - 1 \right]$$

$$\frac{-9}{H} = 0 + A \left[\frac{-1}{H} - 1 \right]$$

$$\frac{-9}{H} = A \left[\frac{-1-H}{H} \right]$$

$$\frac{-9}{H} = A \left[\frac{-1-H}{H} \right]$$

$$\frac{-9}{H} = \frac{-1-H}{H} A$$

$$\frac{-9}{H} = \frac{-1-H}{H} A$$

$$\frac{-9}{H} = \frac{-1-H}{H} A$$

$$B = \frac{9 \times 1}{3} \quad B = 3$$

$$\frac{18z}{(z^2-1)(H^2+1)} = \frac{3}{(z^2-1)} + \frac{3}{(H^2+1)} \rightarrow (2)$$

Using eq (2) in (1).

$$\frac{U(z)}{z} = \frac{3}{(z^2-1)} + \frac{3}{(H^2+1)}$$

$$\frac{U(z)}{z} = \frac{3}{z^2-1} + \frac{3}{H^2+1}$$

$$\frac{Uz}{z} = \frac{3}{2} \frac{1}{[z-1]} + \frac{3}{4} \frac{1}{[z+1/4]}$$

$$Uz = z \left[\frac{3}{2} \left[\frac{1}{(z-1)} \right] + \frac{3}{4} \left[\frac{1}{(z+1/4)} \right] \right]$$

$$Uz = \frac{3}{2} \left[\frac{z}{(z-1)} \right] + \frac{3}{4} \left[\frac{z}{(z+1/4)} \right]$$

Take z inverse on both sides

$$z^{-1}[U(z)] = z^{-1} \left[\frac{3}{2} \left(\frac{z}{z-1} \right) + \frac{3}{4} \left(\frac{z}{z+1/4} \right) \right]$$

$$U_n = \frac{3}{2} z^{-1} \left[\frac{z}{z-1} \right] + \frac{3}{4} z^{-1} \left[\frac{z}{z+1/4} \right]$$

$$U_n = \frac{3}{2} \left(\frac{1}{2} \right)^n + \frac{3}{4} \left(\frac{1}{4} \right)^n$$

→ Applications to z -transforms to difference equation.

$$\Rightarrow U_{n+1} - 3U_n = 4 \quad U_0 = 1$$

$$\text{Given } U_{n+1} - 3U_n = 4$$

Take z transform on both sides

$$z[U_{n+1} - 3U_n] = z[4]$$

$$z[U_{n+1}] - 3z[U_n] = z[4z(1)] \rightarrow \textcircled{1}$$

We know that $z(U_{n+1}) = z[z(U_n) - U_0]$

Using that in eq $\textcircled{1}$.

$$z [2(u_n) - u_0] = 3z [u_n] = \frac{4z}{z-1}$$

$$z [2(u_n) - 1] - 3z [u_n] = \frac{4z}{z-1}$$

$$u_0 = 1 \text{ and } z [u_n] = u_k$$

$$2(u_2) - z - 3z [u_2] = \frac{4z}{z-1}$$

$$(2-3)(u_2) - z = \frac{4z}{z-1}$$

$$(2-3)(u_2) = \frac{4z}{z-1} + z$$

$$(2-3)(u_2) = \frac{4z + z(z-1)}{z-1}$$

$$\cancel{(2-3)}(u_2) = \frac{4z + \cancel{z^2} - z}{\cancel{z}z-1}$$

$$(2-3)(u_2) = \frac{4z + z^2 - z}{z-1}$$

$$(2-3)(u_2) = \frac{\cancel{3z} + z^2 + 3z}{z-1}$$

$$u_2 = \frac{z^2 + 3z}{(z-1)(2-3)}$$

$$u_2 = \frac{z(z+3)}{(z-1)(2-3)}$$

$$\frac{u_2}{z} = \frac{z+3}{(z-1)(2-3)} \rightarrow \textcircled{2}$$

By partial fraction method.

$$\frac{z+3}{(z-1)(z-3)} = \frac{A}{(z-1)} + \frac{B}{(z-3)}$$

$$\frac{z+3}{\cancel{(z-1)(z-3)}} = \frac{A(z-3) + B(z-1)}{\cancel{(z-1)(z-3)}}$$

$$(z+3) = A(z-3) + B(z-1)$$

$$\text{Put } z=3$$

$$3+3 = A(3-3) + B(3-1)$$

$$6 = A(0) + B(2)$$

$$6 = 0 + B(2)$$

$$B = \frac{6}{2} = 3 \quad \therefore \underline{B=3}$$

$$\text{Put } z=1$$

$$1+3 = A(1-3) + B(1-1)$$

$$4 = A(-2) + B(0)$$

$$4 = A(-2)$$

$$A = \frac{4}{-2} \quad \underline{A=-2}$$

$$\therefore \frac{z+3}{(z-1)(z+3)} = \frac{-2}{(z-1)} + \frac{3}{(z-3)} \rightarrow \textcircled{3}$$

using $\textcircled{3}$ in $\textcircled{2}$.

$$\frac{u_2}{z} = \frac{-2}{(z-1)} + \frac{3}{(z-3)}$$

$$\frac{u_2}{z} = -2 \left[\frac{1}{(z-1)} \right] + 3 \left[\frac{1}{z-3} \right]$$

$$u_2 = -2z \left[\frac{1}{z-1} \right] + 3z \left[\frac{1}{z-3} \right]$$

$$U_2 = -2 \left[\frac{z}{z-1} \right] + 3 \left[\frac{z}{z-3} \right]$$

Take inverse on B.S.

$$z^{-1} [U(z)] = z^{-1} \left[-2 \left[\frac{z}{z-1} \right] + 3 \left[\frac{z}{z-3} \right] \right]$$

$$U_n = -2 z^{-1} \left[\frac{z}{z-1} \right] + 3 z^{-1} \left[\frac{z}{z-3} \right]$$

$$z^{-1} \left[\frac{z}{z-a} \right] = a^n$$

$$\therefore U_n = -2(1)^n + 3(3)^n$$

→ Solve $y_{n+2} + 2y_{n+1} + y_n = 0$ with $y_0 = 0, y_1 = 1$

Given $y_{n+2} + 2y_{n+1} + y_n = 0$.
Take z transform on both sides
 $z[y_{n+2} + 2y_{n+1} + y_n] = z(0) \rightarrow \textcircled{1}$

$$\text{Using } z[y_{n+2}] = z[z(y_n) - y_0 - y_1 z^{-1}]$$

and $z[y_{n+1}] = z[z(y_n) - y_0]$ in $\textcircled{1}$ we get.

$$z^2[z(y_n) - y_0 - y_1 z^{-1}] + 2z[z(y_n) - y_0] + z[y_n] = 0$$

$$y_0 = 0, y_1 = 1$$

$$z^0 [y(z) - (0) - (1) \frac{1}{z}] + 2z [y(z) - (0)] + y(z) = 0$$

$$z^2 y(z) - z + 2z y(z) + y(z) = 0.$$

$$(z^2 + 2z + 1) y(z) - z = 0.$$

$$(z^2 + 2z + 1) y(z) = z$$

$$y(z) = \frac{z}{(z^2 + 2z + 1)} = \frac{z}{(z+1)^2}$$

Take inverse on B.S.

$$z^{-1} [y(z)] = z^{-1} \left[\frac{z}{(z+1)^2} \right] \quad z^{-1} \left[\frac{z}{(z+1)^2} \right]$$

$$z^{-1} [y(z)] =$$

$$y_n = z^{-1} \left[\frac{z}{(z+1)^2} \right]$$

We know that $z^{-1} \left[\frac{az}{(z-a)^2} \right] = a^n n$

$$y_n = \left(\frac{-1}{-1} \right) z^{-1} \left[\frac{z}{(z - (-1))^2} \right]$$

$$y_n = -z^{-1} \left[\frac{-1z}{(z - (-1))^2} \right]$$

$$y_n = -(-1)^n n = \underline{\underline{(-1)^{n+1} n}}$$

$$3) \quad u_{n+2} + 4u_{n+1} + 3u_n = 3^n.$$

$$u_0 = 0$$

$$u_1 = 1$$

$$\text{Given } u_{n+2} + 4u_{n+1} + 3u_n = 3^n.$$

Take z transform on both sides

$$z[u_{n+2} + 4u_{n+1} + 3u_n] = z[3^n] \rightarrow \text{①}$$

We know that

$$z(u_{n+2}) = z^2[2(u_n) + u_0 - u_1 z^{-1}]$$

$$z(u_{n+1}) = z(2(u_n) - u_0)$$

$$= z^2[2(u_n) + u_0 + u_1 z^{-1}] + 3z(u_n) = \frac{4z}{z-1}$$

$$u_0 = 0 \text{ and } z$$

$$= z^2[2(u_n) + u_0 + u_1 z^{-1}] + 4z[2(u_n) - u_0] + 3z(u_n) = \frac{2}{z-3}$$

$$u_0 = 0 \quad u_1 = 1 \text{ and } z[u_n] = u(z)$$

$$z^2[u(z) + 0 + 1z^{-1}] + 4z[u(z) - 0] + 3u(z) = \frac{2}{z-3}$$

$$z^2 u(z) - z + 4zu(z) + 3u(z) = \frac{2}{z-3}$$

$$(z^2 + 4z + 3)u(z) - z = \frac{2}{z-3}$$

$$(z^2 + 4z + 3)u(z) = \frac{z}{z-3} + z$$

$$(z^2 + 4z + 3)u(z) = \frac{z + z(z-3)}{z-3}$$

$$(z^2 + 4z + 3)u(z) = \frac{z^2 - 2z}{z-3}$$

$$(z^2 + 4z + 3)U(z) = \frac{z^2 - 2z}{z - 3}$$

$$U(z) = \frac{z^2 - 2z}{(z - 3)(z^2 + 4z + 3)}$$

$$U(z) = \frac{z^2 - 2z}{(z - 3)(z + 3)(z + 1)}$$

$$U(z) = \frac{z(z + 2)}{(z - 3)(z + 3)(z + 1)}$$

$$\frac{U(z)}{z} = \frac{z + 2}{(z - 3)(z + 3)(z + 1)} \longrightarrow \textcircled{2}$$

Using partial fraction method.

$$\frac{z + 2}{(z - 3)(z + 3)(z + 1)} = \frac{A}{(z - 3)} + \frac{B}{(z + 3)} + \frac{C}{(z + 1)}$$

$$\frac{z + 2}{(z - 3)(z + 3)(z + 1)} = \frac{A(z + 3)(z + 1) + B(z - 3)(z + 1) + C(z - 3)(z + 3)}{(z - 3)(z + 3)(z + 1)}$$

$$z + 2 = A(z + 3)(z + 1) + B(z - 3)(z + 1) + C(z - 3)(z + 3)$$

Put $z = -3$

$$-3 + 2 = A(-3 + 3)(-3 + 1) + B(-3 - 3)(-3 + 1) + C(-3 - 3)(-3 + 3)$$

$$-1 = A(0)(-2) + B(-6)(-2) + C(-6)(0)$$

$$-1 = 0 + 12B + 0$$

$$B = -\frac{1}{12}$$

$$\underline{\underline{5}}$$

Put $z = 3$.

$$3 - 2 = A(3+3)(3+1) + B(3-3)(3+1) + C(3+3)(3+3)$$

$$1 = A(6)(4) + B(0)(4) + C(6)(6)$$

$$1 = 24A + 0 + 0$$

$$= A = \frac{1}{24}$$

Put $z = -1$, $-3 = 0 + 0 - 8C = C = \frac{3}{8}$

$$\therefore \frac{z-2}{(z-3)(z+3)(z+1)} = \frac{1/24}{(z-3)} + \frac{5/12}{(z+3)} + \frac{3/8}{(z+1)} \quad \text{--- (2)}$$

Using (3) in (2)

$$\frac{Uz}{z} = \frac{1/24}{(z-3)} + \frac{5/12}{(z+3)} + \frac{3/8}{(z+1)}$$

$$Uz = z \left[\frac{1/24}{(z-3)} + \frac{5/12}{(z+3)} + \frac{3/8}{(z+1)} \right]$$

$$Uz = \frac{1}{24} \left[\frac{z}{z-3} \right] + \frac{5}{12} \left[\frac{z}{z+3} \right] + \frac{3}{8} \left[\frac{z}{z+1} \right]$$

Take inverse on both sides.

$$z^{-1}(Uz) = z^{-1} \left[\frac{1}{24} \left(\frac{z}{z-3} \right) + \frac{5}{12} \left(\frac{z}{z+3} \right) + \frac{3}{8} \left(\frac{z}{z+1} \right) \right]$$

$$U_n = \frac{1}{24} z^{-1} \left[\frac{z}{z-3} \right] + \frac{5}{12} z^{-1} \left[\frac{z}{z+3} \right] + \frac{3}{8} z^{-1} \left[\frac{z}{z+1} \right]$$

$$z^{-1} \left[\frac{z}{z-a} \right] = a^n \quad \text{and} \quad z^{-1} \left[\frac{z}{z+a} \right] = -a^n.$$

$$\therefore U_n = \frac{1}{24} (3)^n + \frac{5}{12} (-3)^n + \frac{3}{8} (-1)^n$$